The Journal of Modern African Studies, 21, 3 (1983), pp. 535–543

The Tanzania-Zambia Railway: the Chinese Loan and the Pre-Investment Analysis Revisited

by Ngila Mwase, *Senior Research Fellow, Economic Research Bureau, University of Dar es Salaam*

The idea of a railway from the Zambian Copperbelt to the Indian Ocean via Tanzania had long been considered. But although the first serious report was published in 1952 by a consultant from the British Colonial Office,[1] it was not until 1964, the year of independence, that further studies were commissioned, all of which eventually found such a railway uneconomic. The World Bank mission erred in tying the cost estimates to the trans-shipment of goods (due to gauge differences), and ignoring the option (actually taken) of constructing a direct line to Dar es Salaam.[2] The U.N. team argued that the reasons for the project were too speculative, that the existing railways had spare capacity, and that the likely development benefits were invalid, unless an iron and steel complex emerged.[3] In 1965 the authors of the Brookings Institution study felt that the project was an unnecessary duplication of existing facilities, and that transport matters in the region should be depoliticised,[4] an idealistic proposal since there are external benefits and costs involved in the operation of railways. Moreover, the private interests involved – in this case, the owners, Rhodesian Railways – can and do impose 'non-economic', albeit monopolistic sanctions. The Stanford Research Institute also produced a negative report in 1966.[5]

The basic assumption of these post-independence analysts was that Zambia would continue to use the Rhodesia Railways for its copper exports. To them the issue was one of competing alternatives – hence, the willingness of both the World Bank and the U.S. Agency for International Development to finance instead the Tanzania-Zambia highway.[6] As far as the Governments of Julius Nyerere and Kenneth Kaunda were concerned, however, they did not consider the two modes of transportation to be mutually exclusive. Both wanted the new railway to end Zambia's reliance on the Rhodesian and Angolan routes by providing a safe, cheap, and permanent access to the sea through a friendly country, at the same time enabling Tanzania to exploit her then-inaccessible Southern Highlands. In 1966, a British-Canadian feasibility study organised by Maxwell Stamp Associates assumed that almost all of Zambia's copper and most of her foreign trade would be carried on the proposed line, and hence came to the conclusion that it was economically feasible, even if other

[1] United Kingdom, Colonial Office, *Report on Central African Railway Link Development Survey* (London, 1952), Vols. 1 and 2.

[2] World Bank, *Report on the North East Rail Link. Summary and Conclusions* (Lusaka, 1964).

[3] *Report of the UN/ECA/FAO Economic Survey Mission on the Economic Development of Zambia* (Ndola, 1964), led by Dudley Seers.

[4] E. Haefele and E. Steinberg, *Government Controls on Transport: an African case* (Washington, D.C., 1965), under the auspices of the Brookings Institution.

[5] Stanford Research Institute, *Tanzania-Zambia Highway Study* (Menlo Park, California, 1966).

[6] The Tanzania-Zambia Railway must have appeared as a threat to the World Bank because of its heavy investment in the East African Railways, and the attached conditions that barred loans for alternative lines, particularly in the direction of Zambia.

https://doi.org/10.1017/S0022278X00023557 Published online by Cambridge University Press

TABLE 1

Comparative Cost Estimates for the Tanzania-Zambia Railway[1]

million Tanzanian Shs.[a]

	World Bank 1963	East African Railways & Harbours 1964	Maxwell Stamp Associates 1966
Cost of new construction	807	748	1,378
Conversion of Tanzania's central line	—	—	196
Contingencies	162	—	174
Rolling stock	60	107	530
Interest during construction[b]	140	171	178
Working capital	—	—	70
Total	1,169	1,026	2,526

a Conversions made at the rate of 1 Zambian Kwacha = 10 Tanzanian Shs.

b The 1963 estimates by the World Bank did not specify the rate of interest, but those for 1964 and 1966 proposed 7% and 6%, respectively.

alternative outlets to the sea could be used.[2] Despite the positive findings of the two-volume report, however, Britain, the United States, and the U.S.S.R., as well as the World Bank and some multinational companies, including Lonrho, rejected requests to assist in the construction of this project. In 1967 the Governments of Zambia and Tanzania concluded an agreement with China to build the railway.[3]

The big differences in the various cost estimates are reflected in Table 1, and can be explained by the dissimilar techniques applied. The study by Maxwell Stamp, for example, was the most detailed, and took the project up to the final design stage. Other variations were a function of time and inflation: the estimated costs were raised by the World Bank by about £11 million in one year, by Maxwell Stamp by as much as £75 million in two years, while the Chinese increased their figures in four years by nearly £45 million.

The Maxwell Stamp Report

The detailed study by Maxwell Stamp Associates deserves most attention, not only since this unequivocally found the project to be economically sound, but also because although the Chinese undertook a number of engineering and

[1] Source: adapted from R. M. Bostock, 'The Transport Sector', in Elliott (ed.), op. cit. p. 368.

[2] Maxwell Stamp Associates et al., 'A British-Canadian Report on an Engineering and Economic Feasibility Study for a Proposed Zambia-East Africa Rail Link', Vols. 1 and 2, London, 1966.

[3] See George Yu, 'Working on the Railroad: China and the Tanzania-Zambia Railway', in *Asian Survey, 1971* (Hong Kong), pp. 1101–17; Charles Elliott (ed.), *Constraints on the Economic Development of Zambia* (Nairobi and London, 1971); Richard Hall and Richard Peyman, *The Great Uhuru Railway: China's showpiece in Africa* (London, 1976); and Kasuka S. Mutukwa, *Politics of the Tanzania-Zambia Rail Project: a study of Tanzania-China-Zambia relations* (Washington, D.C., 1977).

https://doi.org/10.1017/S0022278X00023557 Published online by Cambridge University Press

TABLE 2

Financial Forecasts for the Tanzania-Zambia Railway, 1971–95[1]

	1971	1981	1995
1. Traffic ('000 tons)	2,554	4,281	5,649
2. Operating costs (£ million)			
Survey rate	10·8	17·5	23·1
Rhodesia Railways rate	14·9	24·1	31·7
3. Operating surplus (£ million)			
Survey rate	11·4	18·5	24·2
Rhodesia Railways rate	7·3	11·9	15·6
4. Capital employed (£ million)	126·5	142·7	156·5
5. Gross rate of return (%)			
Survey rate	9·0	13·0	15·3
Rhodesia Railways rate	5·7	8·3	10·0
6. Interest @ 6% on item 4, less £3·5 million working capital	7·4	8·4	9·2
7. Net surplus (3–6) (£ million)			
Survey rate	4·0	10·1	15·0
Rhodesia Railways rate	−0·1	3·5	6·6

costing studies, they assumed prior economic feasibility based on the detailed British-Canadian report. Furthermore, as Table 1 shows, not only is the gross estimate of costs by the Chinese remarkably close to that by Maxwell Stamp, but many of the technical, and especially engineering, recommendations were quite similar.

The importance of these figures is heightened by the fact that the Chinese have yet to produce, let alone publish, their detailed cost-breakdowns. Even today, the Chinese continue to give total figures for incoming bulk railway equipment rather than costing item by item, as is usually the case with other contracting parties. The capital costs were estimated to be £126.5 million in 1971, of which construction amounted to £96.3 million, locomotives and rolling stock to £26.5 million, and working capital to £3.5 million.

Unlike previous reports, Maxwell Stamp Associates estimated that the Rhodesia Railways would not suffer, and that investments to the tune of £84.1 million would have to be made to cope with the expansion of Zambian traffic if the railway to Tanzania was not built. They therefore considered that incremental investments of £12.2 million were well worthwhile, although if locomotives, rolling stock, and working capital were included, this figure would rise to as much as £42.2 million.[2]

It was calculated that operating costs would be much lower than on the Rhodesia Railways. Taking cognizance of the estimated traffic and average

[1] Source: Maxwell Stamp Associates, op. cit. Vol. 1, p. 293.

[2] One of Maxwell Stamp's consultants has argued that even if political considerations were set aside, the Tanzania-Zambia project could offer an alternative to double tracking the single line on the Rhodesia Railways (of which Zambia was a joint owner), since the costs were comparable. See E. Harrison, 'Reorientation of Transportation Patterns from Zambia after Rhodesian UDI', Ph.D. dissertation, Columbia University, New York, 1972.

https://doi.org/10.1017/S0022278X00023557 Published online by Cambridge University Press

haul, the consultants subtracted the operating costs from the expected revenue, and then divided the surplus by the capital costs to get the gross rate of return on the capital employed. They forecast that this would rise from 9 per cent in 1971, to 13 per cent in 1981, and to 15.3 per cent in 1995. They assumed further an interest rate of 6 per cent per annum, leaving a small net surplus estimated to range from £4 million in 1971 to £15 million in 1995, with an initial loss of £ − 0.1 million at the Rhodesia Railways rate. Thus, economic viability was established, albeit mainly on the basis of two operating cost-rates.

A full-scale economic analysis – perhaps employing shadow prices – ought to have been used to supplement any financial appraisal. Maxwell Stamp's high rates of return on capital were unwarranted, because railways – and, indeed, most transport investments – are classic cases of public activity that involve social costs and benefits. Rarely can any transport mode recoup all costs. The average rate of return on fixed assets for the East African Railways, for example, was 6 per cent, which corresponds closely with that realised elsewhere in the world. In general, although 20–30 years is generally adopted as the life span of transport projects in Africa, financiers less generous than the Chinese often prefer a quick return and hence a shorter pay-back period.

This emphasis on a social cost-benefit analysis does not rule out the profitability criterion. Indeed, as A. C. Harberger has observed, rail charges

> constitute direct devices by which the costs of the relevant facilities can, over time, be recouped from the beneficiaries. To a first approximation, therefore, the worthwhileness of such projects can be judged by the strictly commercial criterion of prospective profitability.[1]

This, of course, only applies to the direct costs and benefits of a private project. Maxwell Stamp's appraisal in Table 2 can therefore be considered as a 'first approximation', since other expected indirect benefits of a developmental and political nature further rationalised the project. By way of contrast, most of the other feasibility studies ignored the political logic of Zambia's transport initiatives after the 1965 unilateral declaration of independence in Rhodesia.

Although the problem of choosing between partly conflicting objectives is fundamental to economics, its practical implications have only recently begun to concern economic planners. As S. Marglin has noted, policy-makers find it very difficult, if not impossible, to define and assign weights to conflicting objectives in order to resolve decisions.[2] This, in fact, raises broad questions about the degree to which technical studies can relate to matters of an economic and political nature. Two dominant strategies are available in the literature: (i) the 'decision-making approach' advocated by U.N.I.D.O. and Little/Mirrlees,[3] whereby the appraisal is made in the light of objectives chosen by those who will eventually decide what is to be done, and (ii) the 'Paretian approach' advocated by E. Mishan,[4] whereby the analyst works independently of the political decision-making process by using his own independent norms,

[1] A. C. Harberger, 'Cost-Benefit Analysis of Transportation Projects', in his *Project Evaluation* (London, 1972), p. 248.

[2] S. Marglin, 'The Social Rate of Discount and Optimal Rate of Investment', in *Quarterly Journal of Economics* (Cambridge, Mass.), LXXVII, 1, 1963, pp. 95–111.

[3] U.N.I.D.O., *Guidelines for Project Evaluation* (New York, 1972), and I. M. D. Little and J. A. Mirrlees, *Project Appraisal and Planning for Developing Countries* (London, 1974).

[4] E. Mishan, *Elements of Cost-Benefit Analysis* (London, 1972).

https://doi.org/10.1017/S0022278X00023557 Published online by Cambridge University Press

usually, including narrow considerations of 'economic efficiency' and 'distributional justice'. Thus, unlike the Roskill Commission on the Third London Airport,[1] which to some extent used technical studies to justify political factors, these were discounted by the appraisals of the Tanzania-Zambia railway. A political economy approach was, perhaps, needed to modify the purely 'economic' analysis of the consultants, although their arguments regain some credence in the current changed situation.

The Chinese Loan

The contract for the Tanzania-Zambia railway was Shs. 2,865 million, of which Shs. 2,300 million was an interest-free loan, to be utilised over the 1968–77 decade, followed by a five-year period of grace, plus a long pay-back period, beginning in 1983 and ending in the year 2,001. An estimated 48 per cent of the total cost was for equipment and technical services, of which 16 per cent was for rolling stock. The local costs amounted to 52 per cent of the total (including the extremely modest figure of £28,000 for the Chinese technicians), and was to be met from the proceeds of the sales in Tanzania and Zambia of Chinese goods under a commodity credit agreement, estimated to realise annually about £8.5 million over five years for each country.

Although the Chinese contract was generous in some respects, the removal of certain 'undesirable' features could have brought the loan closer to most international terms.[2] For example, the inclusion in the contract, not only of construction costs, but also rolling stock, locomotives, signalling equipment, etcetera, meant that even if the Chinese could not supply these, they were allowed to decide who to buy from, and on what terms. According to M. Weil, some second-rate equipment for construction was bought.[3] Be that as it may, for purposes of compatibility, the railway has to rely heavily on Chinese materials and equipment, which have less standard parts than is normally the case.[4]

It would have been more economical to have *borrowed* the funds for the local costs that were tied to Chinese imports, some of which were relatively expensive, inconvenient, and competitive as regards local manufactures. These imports rose from about £4 million – of which £2.9 million was for consumer goods – in 1969 to about £11 million per year in the early 1970s. Both Tanzania and Zambia were hard pressed to find suitable Chinese imports in a short space of five years, partly because local industries could not survive

[1] [Roskill] *Commission on the Third London Airport. Papers and Proceedings. Stage III, Research and Investigation–Assessment of Short-Listed Sites* (London, 1970), Vol. 7, pts. 1 and 2.

[2] Some I.D.A. loans to Tanzania road projects, for example, have been free of interest, payable over 50 years after a 10-year grace period, except for a commitment charge of ¾%. Countries giving such 'soft loans' include Canada, Sweden, the Netherlands, and the United Kingdom, and their Governments have sometimes agreed to 'write off' these debts. See *Speech by the Minister for Finance, Introducing the Estimates of Public Revenues and Expenditure to the National Assembly* (Dar es Salaam, 1972), para. 86.

[3] M. Weil, 'The TANZAM Railway: a case study in project development', M.A. thesis, University of Sussex, Brighton, 1972.

[4] The problem is exacerbated by the fact that the Chinese priced their inputs into the Tanzania-Zambia railway in bulk – e.g. one shipload. Even the railway's actual cost was only settled after the project had been completed. Thus, more trust was required of Tanzania and Zambia than was needed with most western bilateral or international aid agencies.

https://doi.org/10.1017/S0022278X00023557 Published online by Cambridge University Press

TABLE 3

Source of Tanzania's Imports of Certain Products, 1969 and 1973[1]

million Shs.

	China		Kenya	
	1969	1973	1969	1973
Dentifrices	—	8·6	3·2	0·1
Toilet soap	—	10·1	9·2	2·0
Detergents	—	9·2	8·2	—
Cement	1·3	14·2	14·1	2·5
Shirts	—	4·6	1·6	0·1

cheap Chinese imports of textiles, bicycles, chinaware, pharmaceuticals, toothpaste, etcetera, although some met strong consumer resistance and remained largely unsold. In 1964, Tanzanian imports from China had been a mere £0.32 million, about 0.7 per cent of the total, excluding East African inter-state trade. Imports rose drastically with the commodity credit agreement: whereas in 1969 China supplied less than 5 per cent of Tanzania's total imports, by 1973 the proportion had risen to 22.4 per cent,[2] compared with less than 1 per cent from both Kenya and Uganda. Over the same period, Tanzania's imports from her Common Market partners fell from 17 per cent to less than 10 per cent. Table 3 suggests that the increase in Chinese imports was at the expense of goods from the rest of East Africa – as witnessed by the decline in imports from Kenya.

Some Chinese imports were so highly competitive in price that a long-term shift in the pattern of Tanzanian trade might have taken place at the expense of East Africa[3] – indeed, there was a real danger that cheap Chinese goods might be 'dumped'. Thus, far from the loan being 'without strings', it was directly tied to trade. However, despite George Yu's assertion that the commodity credit agreement offset China's foreign-exchange scarcities,[4] R. Makoni was perhaps correct in concluding that this was unlikely to be any worse than the monetary arrangements on equivalent western loans.[5] In any

[1] Source: Arthur Hazlewood, *Economic Integration: the East African experience* (London, 1975), p. 111.

[2] The increase in imports from China was related more to the railway project than to the diversification of trade, as may be seen from the fact that the value of Tanzania's imports from China dropped to 7.4 per cent in 1976. Exports to China have been static, rising from 3.3 per cent of total exports in 1964 to a peak of 4.4 per cent in 1973, and stabilising at 3 per cent in 1976.

[3] The adverse impact on East African trade should not be exaggerated. The change in the level of potential trade within the Common Market on account of products imported from China, but also produced in East Africa in 1970–1, for example, was no more than Shs. 18.2 million. However, although these 'barter arrangements' were not in the spirit of the Common Market, the terms of alternative supplies from Kenya were not particularly advantageous. See Common Market and Economic Affairs Secretariat, 'A Report of the Working Party on Possibilities for Closer Harmonization of Monetary/Fiscal and Payments Policies within the East African Community', Arusha, 1973.

[4] Yu, loc. cit.

[5] R. Makoni, 'The Economic Appraisal of the Tanzania-Zambia Railway', in *The African Review* (Dar es Salaam), 2, 4, June 1972, pp. 596–616.

https://doi.org/10.1017/S0022278X00023557 Published online by Cambridge University Press

TABLE 4

Comparative Analysis of Loan-Cost Repayments for the Tanzania-Zambia Railway

Loan (principal) (£m)	Assumed interest rate (%)	Annual repayments 1983–2001 (£m)	Total repayments by 2001 (£m)	Column 4 minus Column 1	
				Difference (£m)	(%)
167·00	0	9·28	167·00	0	0
167·00	1	11·82	212·82	45·82	27·44
167·00	2	14·99	269·86	102·86	61·59
167·00	3	18·92	340·51	173·51	103·89
167·00	4	23·76	427·64	260·64	156·07
167·00	5	29·70	534·60	367·60	220·12
167·00	6	36·96	665·34	498·34	298·41

case, most western donors would have demanded repayment in hard currencies. Indeed, more than 70 per cent of western lenders tie the borrower to the purchase of goods from their own countries. Neither commodity nor monetary agreements are necessarily the best and cheapest.

However, the Chinese contract ought really to be judged by its very generous interest-free loan. This can be demonstrated by considering what the repayments would be if the interest rates varied from 1 to 6 per cent – the latter figure being employed by both the parallel pipeline and highway projects, as well as by the Maxwell Stamp consultant for the railway. It has been assumed that repayments will be made during the 18-year period from 1983 onwards, and then be completed by 2001. Taking the loan as a mortgage, we have an annuity-type situation in which a sum of money is to be repaid with interest by certain payments for a fixed number of years, to be calculated by using the following formula:

$$AR = PV\left[\frac{1-(1+i)^{-N}}{i}\right]^{-1} \qquad (1)$$

where:

AR = annual repayments
PV = present value
i = interest rate
N = number of years.

The 'annual equivalents' formula takes into account the successive reduction of the original debt. In addition, the total annual repayments can be summed up as:

$$TAR = N \times AR. \qquad (2)$$

The results of these calculations are presented in Table 4, where it will be seen that repayments at the rate of 6 per cent would have been almost 4 times that for a zero interest rate. This is backed up by the huge repayment differences

https://doi.org/10.1017/S0022278X00023557 Published online by Cambridge University Press

TABLE 5

Effects of Devaluation on the Tanzania–Zambia Railway Loan

	1968	1975	1977	1978	% change 1968–78
Exchange rate[a] (Shs./Yuan)	2·90	3·70	4·38	4·81	+65·86
Debt in Yuan (millions)	988	988	988	988	—
Debt in Shs. (millions)	2,865·2	3,655·6	4,327·44	4,752·28	+65·86

[a] The 1968 exchange rate for the currencies has been taken from the loan agreement; the other years have been calculated using the I.M.F.-designated rates against the U.S. $.

in the first two columns, both as regards costs and associated percentage differences. Huge hypothetical 'savings' were realised, therefore, from the no-interest nature of this loan.

The railway loan was fixed in Yuan, the Chinese currency, with repayments in the 'hard currencies' used by Tanzania and Zambia, and therefore dependent on international rates of exchange. The effects of the devaluation of the Tanzanian Shilling in 1971 and 1973, for example, are therefore of great interest, as may be seen from Table 5. Indeed, since the loan is shared equally by the two countries, it is assumed that both the Tanzanian Shilling and the Zambian Kwacha have maintained similar relations with the Yuan, i.e. that they have depreciated at the same rate.

The debt in Tanzanian Shillings for the different years is a product of the amount of the fixed loan and the fluctuating exchange rate. Whereas one Yuan was equivalent to Shs. 2.90 in 1968, Table 5 shows that this rose to Shs. 4.81 in 1978, and that the amount of the loan, which stood at Shs. 2,865.2 million in 1968, had nearly doubled a decade later. In fact, a mere one per cent devaluation of the Tanzanian Shilling in 1978 had the effect of 'losing' Shs. 424.84 million in terms of extra loan repayments.[1]

With the fluidity in the international monetary situation and the vulnerability of low-income dependent economies, the loan for both Tanzania and Zambia could quadruple if the currency movements continue in an unfavourable direction. During the period, 1968–78, the fall of the Tanzanian Shilling against the U.S. $ was accompanied by an appreciation of the Chinese Yuan. The latter development, which could not be forecast in 1968 – since the Yuan was then a 'soft' L.D.C.-type currency, virtually detached from world trade – has reduced the 'grant' element of the loan.

Since the problem has emerged because the Yuan has risen against the hard currencies in which the repayment is due, it can be seen with hindsight that the best arrangement would have been if this had been fixed in Shillings or Kwachas. Since Tanzania has not been running a balance-of-payments surplus

[1] The Tanzanian Shilling was devalued by 10 per cent in 1982, and by a further 20 per cent in June 1983, in accordance with the demands of the International Monetary Fund.

https://doi.org/10.1017/S0022278X00023557 Published online by Cambridge University Press

vis-à-vis China, and could not therefore pay in Yuan, it would have been better if the repayment had been fixed in U.S. $. By setting the loan in the Yuan equivalent in hard currencies at repayment, rather than at the time of receipt, the 'grant' element forfeited due to currency movements could amount to as much as would be involved in the cumulation of a fixed rate of interest, which could have been payable by alternative international finance.

❋ ❋ ❋

The Tanzania-Zambia railway has often been criticised as a huge and unnecessary project; the traffic potential is shaky, and there is little complimentarity with the expensive and almost parallel road. We can now see that despite the seemingly generous Chinese loan this joint project is proving to be much more costly than had ever been expected.

https://doi.org/10.1017/S0022278X00023557 Published online by Cambridge University Press